Quiz 11 Solutions

Give exact answers and justify your reasoning. Do not use electronic tools. All functions are real-valued, and angles are measured in radians. Whenever you use L'Hôpital's rule, check its hypotheses at each application. Use the strict tangent-line definition of concavity from Lecture 11.

1. Let $f(x) = (x^2 - 1)/(x^2 - x)$ on its natural domain. Identify the excluded points and determine which discontinuities can be removed by assigning a value. Give each required value and find all vertical and horizontal asymptotes.

Solution. The denominator is $x(x-1)$, so the natural domain is $\mathbb{R} \setminus \{0, 1\}$. For x in this domain,

$$f(x) = \frac{(x-1)(x+1)}{x(x-1)} = 1 + \frac{1}{x}.$$

At $x = 1$,

$$\lim_{x \to 1} f(x) = 2.$$

Thus the discontinuity at 1 is removable by assigning $\boxed{f(1) = 2}$. This point is still excluded from the original domain.

At $x = 0$,

$$\lim_{x \to 0^-} f(x) = -\infty, \qquad \lim_{x \to 0^+} f(x) = +\infty.$$

No real value assigned at 0 makes the function continuous there. The only vertical asymptote is $\boxed{x = 0}$: at every other point of the domain the function is continuous, and its limit at 1 is finite.

Finally,

$$\lim_{x \to -\infty} f(x) = \lim_{x \to +\infty} f(x) = 1,$$

so the only horizontal asymptote is $\boxed{y = 1}$, at both infinities.

2. Evaluate

$$\lim_{x\to 0}(\cos x)^{1/x^2}.$$

Solution. For $0 < |x| < \pi/2$, we have $\cos x > 0$, so

$$(\cos x)^{1/x^2} = \exp\left(\frac{\ln(\cos x)}{x^2}\right).$$

The quotient $\ln(\cos x)/x^2$ has form $0/0$. Its numerator and denominator are differentiable on each side of 0, and the denominator derivative $2x$ is nonzero there. Their derivative quotient is

$$\frac{-\tan x}{2x}.$$

This again has form $0/0$, with differentiable numerator and denominator and nonzero denominator derivative 2. Its derivative quotient satisfies

$$\lim_{x\to 0}\frac{-\sec^2 x}{2} = -\frac{1}{2}.$$

Applying L'Hôpital's rule first to the second quotient and then to the first gives

$$\lim_{x\to 0}\frac{-\tan x}{2x} = -\frac{1}{2}, \qquad \lim_{x\to 0}\frac{\ln(\cos x)}{x^2} = -\frac{1}{2}.$$

The same argument applies from both sides. By continuity of the exponential,

$$\boxed{\lim_{x\to 0}(\cos x)^{1/x^2} = e^{-1/2}.}$$